실내공간 무선신호 이변측량을 통한 측위기법

POSITIONING TECHNIQUE THROUGH INDOOR SPACE WIRELESS SIGNAL BILATERATION

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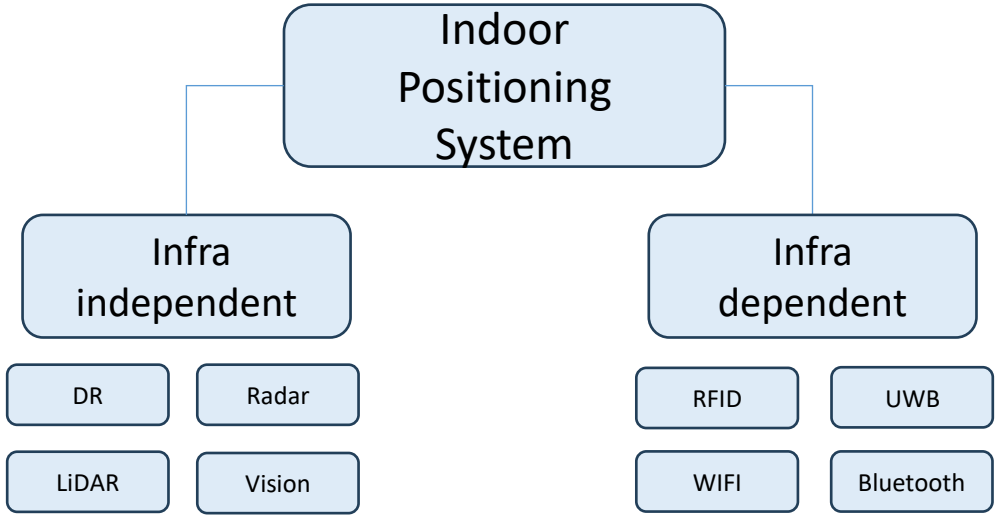
ADVISE PROFESSOR : JINWOO SONG

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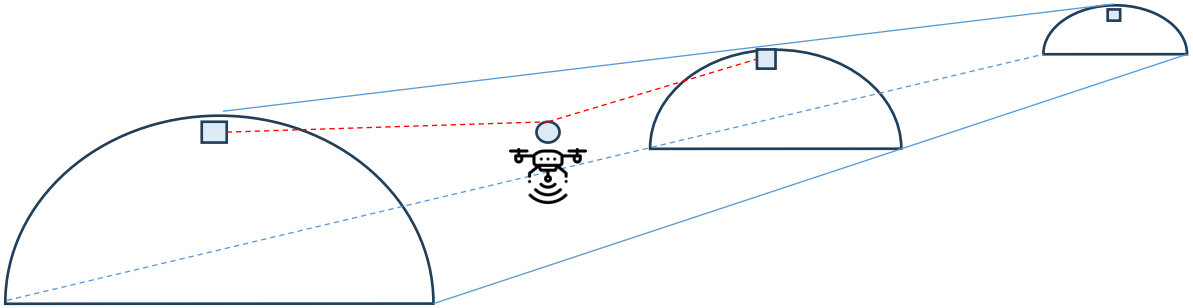
BACKGROUND (1/2)

- As the count of unmanned platforms rises along with the frequency of missions carried out indoors, the demand for indoor positioning escalates.
- Efforts are underway to explore platform positioning techniques within indoor settings through avenues like vision, wireless signals, and map-based approaches.
- Utilizing wireless signals offers benefits in terms of reliability and scalability, while employing UWB (Ultra-Wideband) for positioning can yield highly precise measurements of position.



BACKGROUND (2/2)

- In static or consistently occupied areas, employing the wireless signal-based positioning method can yield accurate measurements by installing four or more anchors.
- Yet, it could prove challenging to deploy multiple anchors in continuous indoor environments or areas with sporadic platform demands.
- Hence, positioning was achieved by cutting down the necessary anchors from four or more to just two.
- And then, a technique for acquiring positioning data via 2d-Lidar and INS information is introduced.



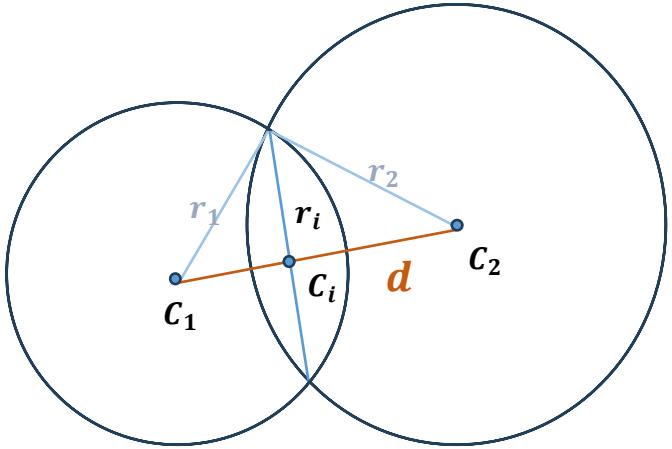
BILATERATION POSITIONING TECHNIQUE(1/5)

- By utilizing the distances measured from the two anchors to the platform node, circle can be determined formed by intersecting spheres.
- To accomplish this, it's important to know the installation coordinates of the two anchors beforehand.
- The center and radius of the resulting circle can be expressed as follows:

$$h = \frac{1}{2} + \frac{r_1 * r_1 + r_2 * r_2}{2d^2}$$

$$\mathbf{c_i} = \mathbf{c_1} + h(\mathbf{c_2} - \mathbf{c_1})$$

$$\mathrm{r_i} = \sqrt{\mathrm{r_1^2} - \mathrm{h^2 d^2}}$$



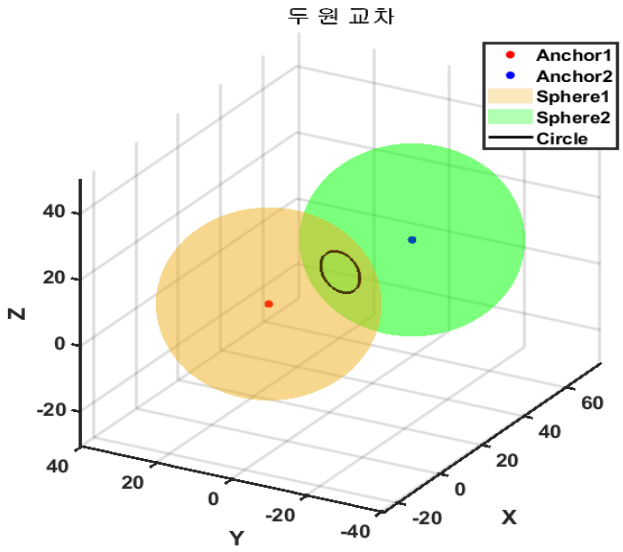
BILATERATION POSITIONING TECHNIQUE(2/5)

- The normal vector of a circle formed where two spheres meet can be described like this:

$$\mathbf{n}_i = (\mathbf{c}_2 - \mathbf{c}_1)/d$$

- The circle can be described by using t_i and b_i, which are directions tangent and bitangent to the normal vector.

$$\boldsymbol{circle}(\boldsymbol{\theta}) = \mathbf{c}_i + \mathrm{r}_i * (\mathbf{t}_i \, cos(\boldsymbol{\theta}) + \mathbf{b}_i \, sin(\boldsymbol{\theta}))$$

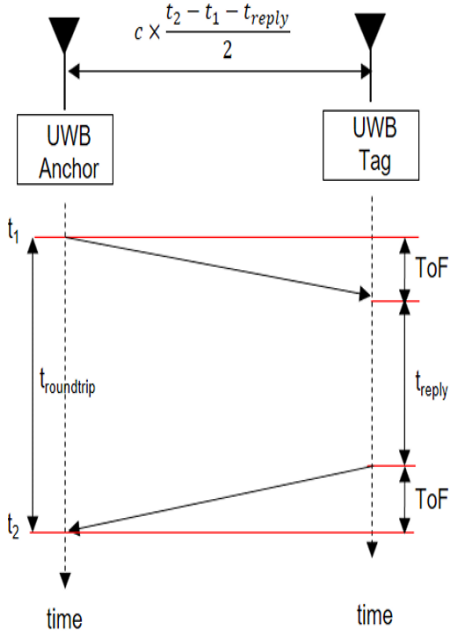


BILATERATION POSITIONING TECHNIQUE(3/5)

- The method of measuring distance using a wireless signal involves calculating the distance by measuring how long the signal takes to travel.
- The distance between an anchor and a node can be expressed using the following formula :

$$range = c * \frac{t_2 - t_1 - t_{reply}}{2}$$

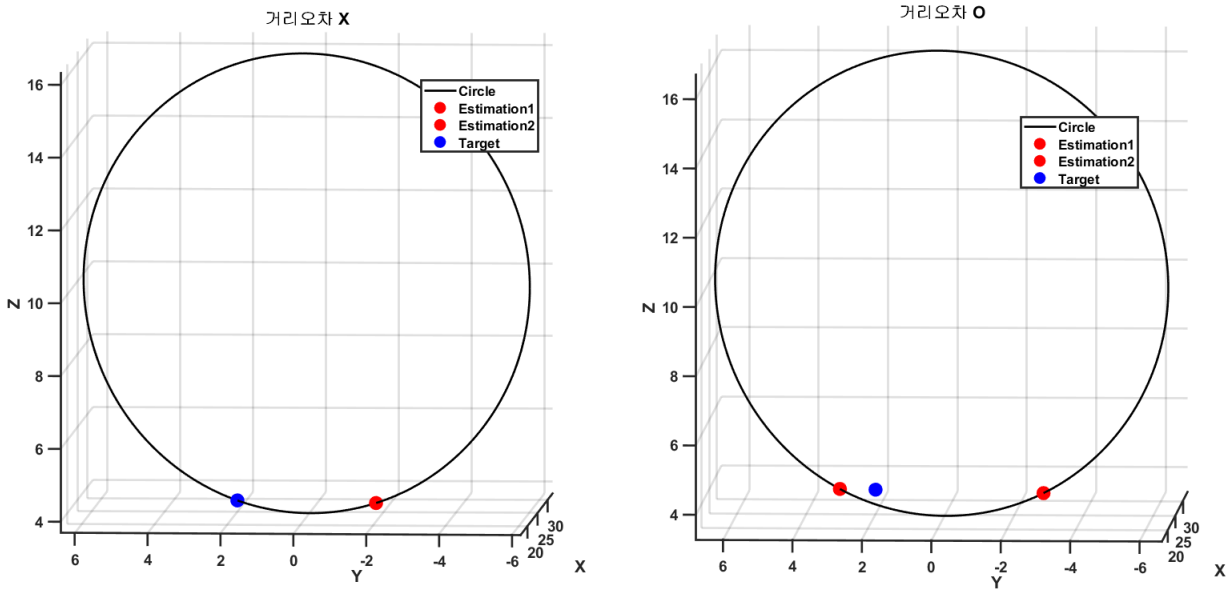
- However, errors in distance information can occur due to clock drift caused by discrepancies in the clocks or signal noise from multipath error
- Referring to previous research, errors were set to increase with distance from the anchor



Distance between anchor and nodes	measurement distance error
minimum error	3cm
maximum error	15cm

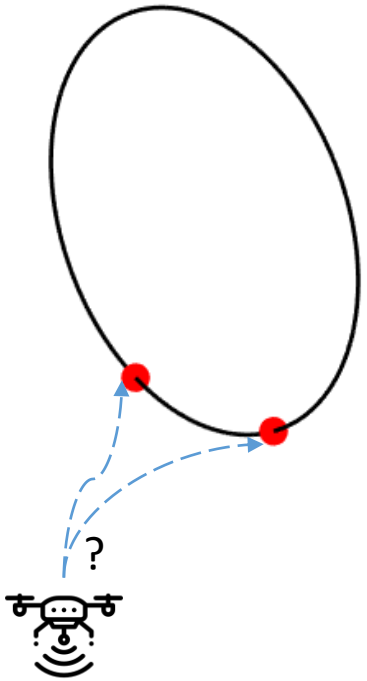
BILATERATION POSITIONING TECHNIQUE(4/5)

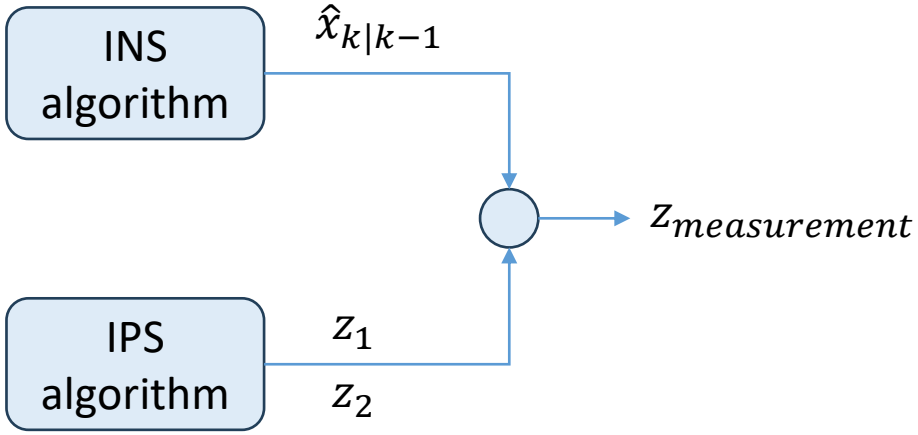
- By using the circle formed by the intersection and the height data measured by the altimeter of the 2D LiDAR attached to the platform, it can identify two points. These points help estimate the platform's position.
- If the wireless signal is error-free, one of these estimated points aligns with the platform's actual location.
- However, if there's an error in the wireless signal, the two estimated points for the platform's actual location may not perfectly align.



BILATERATION POSITIONING TECHNIQUE(5/5)

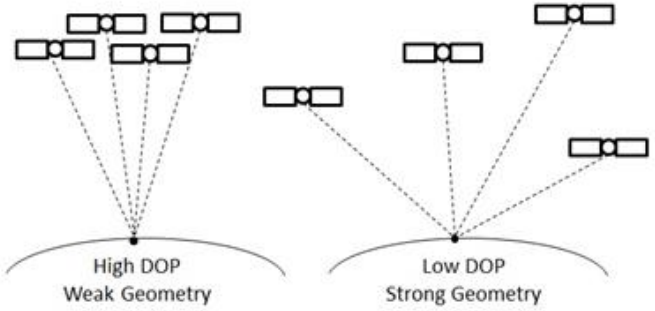
- A process is needed to choose between two estimated points
- Compare the distances between a position estimated using an inertial sensor and a measured value to find the point with the smaller error
- The chosen point is then considered to be the drone's current location and used as the position data in EKF navigation

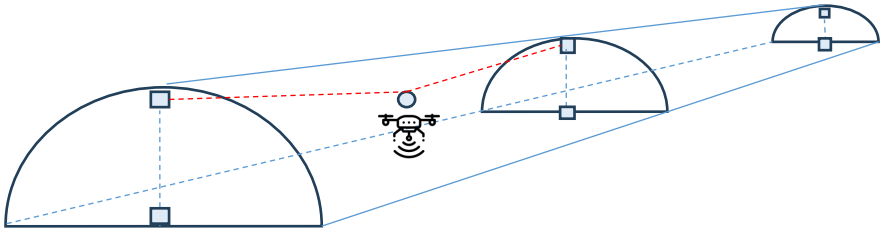




DOP ANALYSIS (1/3)

- DOP measures how good the setup of satellites is when using satellite navigation. A lower DOP value means the measurement is more precise and of better quality
- DOP includes different types like HDOP (horizontal), VDOP (vertical), PDOP (position), TDOP (time), and GDOP (position and time). This study focused on PDOP, which was evaluated based on where anchors were placed
- Since you can't calculate DOP with just 2 anchors, 2 more anchors positioned at right angles to the first two were added. This way, PDOP was calculated using a total of 4 anchors.





DOP ANALYSIS(2/3)

- The process of calculating PDOP can be expressed as follows:

$\boldsymbol{x, y, z}$ = location of receiver

$$\mathbf{R_i} = \sqrt{(x_i - x)^2 + (y_i - y)^2 + (z_i - z)^2}$$

$\boldsymbol{x_i, y_i, z_i}$ = Location of the ith anchor

$\boldsymbol{R_i}$ = Distance between ith anchor and receiver

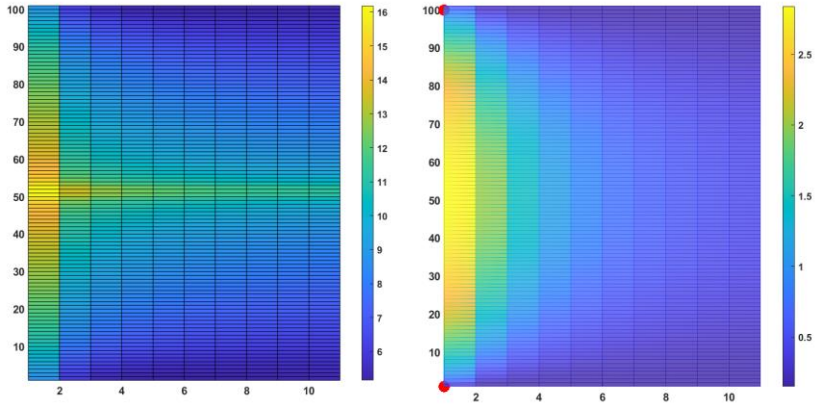
$$\boldsymbol{A} = \begin{bmatrix} \frac{x_1 - x}{R_1} & \frac{y_1 - y}{R_1} & \frac{z_1 - z}{R_1} & 1 \\ \frac{x_2 - x}{R_2} & \frac{y_2 - y}{R_2} & \frac{z_2 - z}{R_2} & 1 \\ \frac{x_3 - x}{R_3} & \frac{y_3 - y}{R_3} & \frac{z_3 - z}{R_3} & 1 \\ \frac{x_4 - x}{R_4} & \frac{y_4 - y}{R_4} & \frac{z_4 - z}{R_4} & 1 \end{bmatrix}$$

$$\boldsymbol{Q} = \left(\boldsymbol{A^T A}\right)^{-1} = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} & \sigma_{xz} & \sigma_{xt} \\ \sigma_{xy} & \sigma_y^2 & \sigma_{yz} & \sigma_{yt} \\ \sigma_{xz} & \sigma_{yz} & \sigma_z^2 & \sigma_{zt} \\ \sigma_{xt} & \sigma_{yt} & \sigma_{zt} & \sigma_t^2 \end{bmatrix}$$

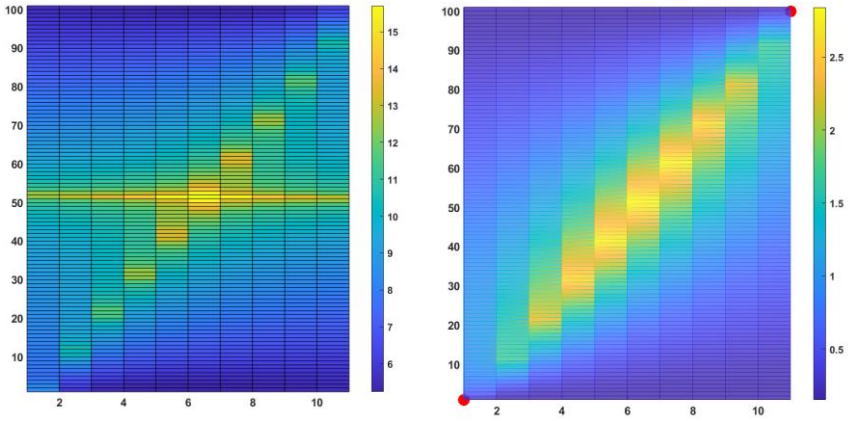
$$PDOP = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2}$$

DOP ANALYSIS (2/3)

- Compare the results of ‘straight installation’ and ‘diagonal installation’ to compare DOP according to the installation positions of the two anchors.
- Because the result from the existing DOP calculation is too large, scaling is performed using the log function.
- It can be observed that the shapes of the DOP calculation map and the measurement error map are similar.
  - Although the DOP map and the measurement map do not completely match, it can explain why the measurement error increases in certain areas.



Straight installation (DOP, position error)

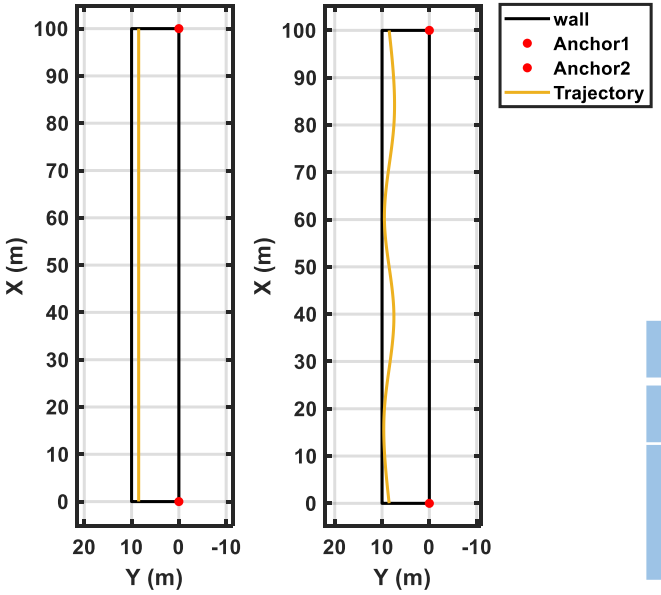


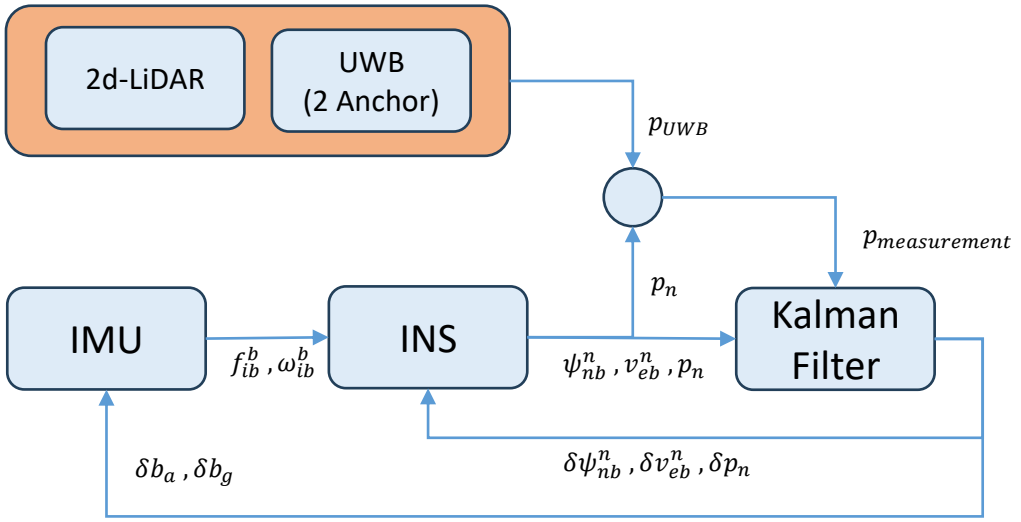
Diagonal installation (DOP, position error)

	straight installation	diagonal installation
Average error (entire section)	77cm	97cm

SIMULATION(1/2)

- Indoor positioning simulation is performed through drone trajectories with ‘line and S-shape’
- Configuring an Extended Kalman Filter (EKF) navigation system that uses position measurements from two anchors.





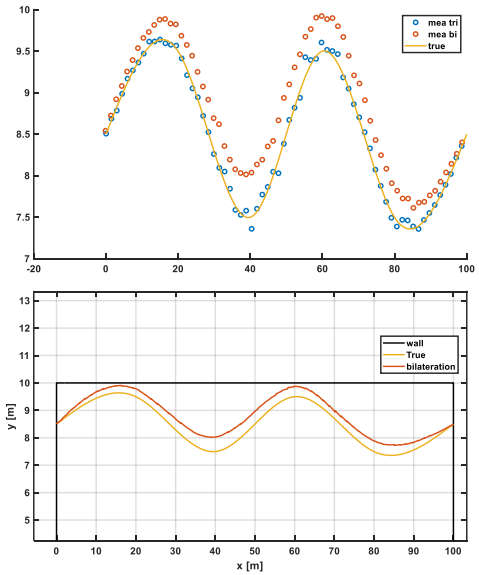
	IMU	UWB	2d-LiDAR
Samplint rate	10hz	2hz	2hz
specification	Accel bias : 50mg gyro bias : 1 rad/sqrt(s)	Clock drift : 100 picosec	Attitude error : 2 degree

SIMULATION(1/2)

- Run 50 Monte Carlo simulations to compare navigation results using bilateration positioning technique.
  - Comparison of true value and position error according to y-axis starting point
  - Error comparison between measurements using four anchors and measurements using two anchors

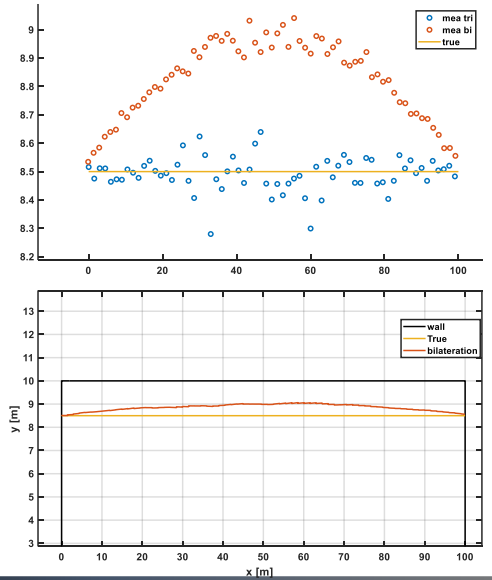
Curve trajectory (start point)	8.5 m	7.5 m	6.5 m
Position error (average)	0.334 m	0.376 m	0.43 m
Measurement error (average)	0.283 m	0.325 m	0.375 m

line trajectory (start point)	8.5 m	7.5 m	6.5 m
Position error (average)	0.335 m	0.35 m	0.41 m
Measurement error (average)	0.28 m	0.317 m	0.365 m



: measurement with 2 anchor

: measurement with 4 anchor



CONCLUSION

- In indoor positioning using wireless signals, location data is typically derived from distance measurements using four anchors.
- In this study, situation when multiple anchors are difficult to install, an attempt was made to obtain distance information using two anchors and location information using a rangefinder and inertial sensor
- DOP analysis confirms that the quality of measurements improves as the anchor moves away from the axis on which it is installed.
- Based on this, through trajectory scenarios, results were obtained within about 50cm at 6.5m in straight and curved trajectories on the installation axis.
- By comparing the given errors in the mobile section and the additional cost of installing two more anchors, it is believed that the method proposed in this study can be used if there are advantages.